Python MySQL Connectivity

Introduction

- **What is MySQL Connectivity?**
  - MySQL is a relational database management system.
  - Python can communicate with MySQL databases using connectors.
  - Used for storing, retrieving, updating, and deleting data.
- **Applications**
  - Student Management System
  - Employee Management System
  - E-Commerce Applications
  - Banking Systems

Requirements

- Install MySQL Connector
  - pip install mysql-connector-python
- Import Module
  - import mysql.connector

Establishing Connection

```
import mysql.connector
conn = mysql.connector.connect(
host="localhost",
user="root",
password="root123",
database="college"
)
print("Connected Successfully")
```

- **Explanation**
  - host → MySQL Server
  - user → Username
  - password → Password
  - database → Database Name

Creating Cursor Object

```
cursor = conn.cursor()
```

- **Why Cursor?**
  - Executes SQL queries.
  - Fetches records from database.
  - Manages database operations.

CREATE Operation

```
cursor.execute("""
CREATE TABLE student(
id INT PRIMARY KEY,
name VARCHAR(50),
course VARCHAR(50)
)""")
```

- **Output**
  - Table created successfully.

Insert Single Record

```
sql = "INSERT INTO student VALUES(%s,%s,%s)"
val = (101,"Amit","Python")
cursor.execute(sql,val)
conn.commit()
print("Record Inserted")
```

- **commit()**
  - Saves changes permanently.

Insert Multiple Records

```
sql = "INSERT INTO student VALUES(%s,%s,%s)"
records = [
(102,"Rahul","Java"),
(103,"Priya","PHP"),
(104,"Ankit","Python")
]
cursor.executemany(sql,records)
conn.commit()
```

- **executemany()**
  - Inserts multiple records at once.

READ Operation

- Select All Records
  - cursor.execute("SELECT * FROM student")

- fetchone(): Retrieves only one row.
- fetchmany() : Retrieves specified number of rows.
- fetchall() :Retrieves all rows from query result.

Example:

➢fetchone()

```
cursor.execute("SELECT * FROM student")
row = cursor.fetchone()
        print(row)
```

Output:

(101, 'Amit', 'Python')

➢fetchmany(int)

```
cursor.execute("SELECT * FROM student")
rows = cursor.fetchmany(2)
for row in rows:
        print(row)
```

Output:

(101,'Amit','Python')
(102,'Rahul','Java')

➢fetchall()

```
cursor.execute("SELECT * FROM student")
rows = cursor.fetchall()
for row in rows:
        print(row)
```

Output:

(101,'Amit','Python')
(102,'Rahul','Java')
(103,'Priya','PHP')
(104,'Ankit','Python')

UPDATE Operation

```
sql = """
UPDATE student
SET course=%s
WHERE id=%s
"""
values = ("Data Science",102)
cursor.execute(sql,values)
conn.commit()
print("Record Updated")
```

DELETE Operation

```
sql = "DELETE FROM student WHERE id=%s"
value = (104,)
cursor.execute(sql,value)
conn.commit()

print("Record Deleted")
```

Complete CRUD Program

```
import mysql.connector
conn = mysql.connector.connect(
host="localhost",
user="root",
password="root123",
database="college"
)
cursor = conn.cursor()


# CREATE
cursor.execute("""
CREATE TABLE IF NOT EXISTS
student(
id INT PRIMARY KEY,
name VARCHAR(50),
course VARCHAR(50)
)
""")
```

```
# INSERT
cursor.execute(
"INSERT INTO student
VALUES(%s,%s,%s)",
(101,"Amit","Python")
)
conn.commit()


# READ
cursor.execute("SELECT * FROM
student")

for row in cursor.fetchall():
print(row)
```

```
# UPDATE
cursor.execute(
"UPDATE student SET course=%s
WHERE id=%s",
("Data Science",101)
)
conn.commit()


# DELETE
cursor.execute(
"DELETE FROM student WHERE
id=%s",
(101,)
)

conn.commit()
cursor.close()
conn.close()
```